Bridge Detection over Water Bodies

Using YOLOv8 Oriented Bounding Box Detection on DOTA v1.0 Aerial Dataset

Yash Dodiya — 230021008 | Arnav Deshpandae — 230021005 | Apoorv Singh — 230021004

Course: AA310 | Framework: Ultralytics YOLOv8 | Platform: Kaggle (Tesla T4 GPU)

Model: YOLOv8s-OBB | Input: 1024×1024 tiles | Training: 100 Epochs | Dataset: DOTA v1.0

Abstract

This report presents a deep learning-based system for automatically detecting bridges over water bodies in high-resolution aerial imagery. Bridges are critical infrastructure elements whose accurate detection and monitoring are essential for urban planning, disaster response, and geographic information systems (GIS). Manual inspection of aerial and satellite imagery is time-consuming and error-prone, motivating the need for automated detection systems.

The project initially explored the GLH-Bridge dataset — a high-resolution bridge-specific dataset containing images ranging from 2,048×2,048 to 16,384×16,384 pixels. However, the extreme image sizes exceeded available GPU memory on Kaggle (Tesla T4) and Google Colab, causing repeated runtime errors. Even after downsampling, the full dataset size of ~50 GB exceeded available disk quotas. The project therefore pivoted to the DOTA v1.0 benchmark, which provides pre-tiled 1024×1024 crops suitable for GPU-constrained training environments.

We leverage YOLOv8's Oriented Bounding Box (OBB) variant trained on DOTA v1.0, which contains 15 object categories including bridges. Two model variants were trained and compared: YOLOv8n-OBB (nano, baseline) and YOLOv8s-OBB (small, improved). The improved model achieved an overall mAP@50 of 54.9% across all 15 DOTA classes, a Bridge-specific AP@50 of 19.2%, Precision of 78.3%, and Recall of 52.8%. These results demonstrate a complete machine learning pipeline: dataset acquisition, YAML configuration, augmentation strategy, training, evaluation, per-class analysis, and deployable inference.

Table of Contents

1. Introduction & Problem Statement

Bridges are among the most critical elements of transportation infrastructure. Their accurate location and condition monitoring is vital for national defence mapping, disaster response planning, flood-risk assessment, and GIS database maintenance. The manual identification of bridges from aerial or satellite imagery is laborious, requiring expert human annotators to scan thousands of square kilometres of imagery.

This project addresses the problem of automating bridge detection from overhead imagery using modern deep learning techniques. Bridges present unique challenges compared to other aerial objects: they are elongated, thin structures that can appear at any orientation, they are visually similar to roads at low resolution, and they represent a rare class in most aerial datasets.

We use YOLOv8-OBB (Oriented Bounding Box) which regresses rotated bounding boxes aligned to the object's true orientation — making it ideally suited to elongated, directionally variable objects like bridges.

2. Dataset Journey: GLH-Bridge → DOTA v1.0

The dataset selection process itself forms an important part of this project's methodology. The initial plan was to use the **GLH-Bridge dataset** — a bridge-specific remote sensing dataset designed to challenge detection systems with very high-resolution imagery.

2.1 GLH-Bridge Dataset — Initial Choice

The GLH-Bridge dataset comprises 6,000 high-resolution bridge images with image sizes ranging from **2,048×2,048 to 16,384×16,384 pixels**. It was selected for its bridge-specific focus and the richness of its annotations. However, attempting to train with this dataset revealed three compounding practical barriers:

Challenge 1 — GPU Memory (OOM Errors)	Challenge 2 — Disk Space Exceeded	Challenge 3 — Preprocessing Bottleneck
Images up to 16,384×16,384 px cannot fit in VRAM even at batch size 1 on a Kaggle Tesla T4 (16 GB) or Google Colab T4. Runtime CUDA out-of-memory errors halted training repeatedly even after aggressive downsampling attempts.	The full GLH-Bridge dataset is approximately **50 GB**. Kaggle notebooks provide ~20 GB of working disk space and Colab even less. Even with symlinks and selective loading, disk quota was exceeded during dataset preparation.	Downsampling 16K images to 1024×1024 is computationally expensive and risks losing the fine structural detail that makes bridge detection meaningful. Any tile-based cropping strategy on images this large requires significant preprocessing infrastructure beyond the scope of a course project.

■ *Due to GPU runtime errors (CUDA OOM), disk space exhaustion (~50 GB dataset), and the impracticality of downsampling 16K-pixel images without losing structural detail, the GLH-Bridge dataset was deemed infeasible for this GPU-constrained environment. The project pivoted to DOTA v1.0, which provides GPU-ready pre-tiled 1024×1024 crops.*

2.2 Pivot to DOTA v1.0 — Rationale

DOTA v1.0 was chosen as the replacement for the following reasons:

- **Pre-tiled format:** Ultralytics distributes DOTA v1.0 as pre-tiled 1024×1024 px crops, eliminating the memory and preprocessing problems encountered with GLH-Bridge.

- **Manageable size:** The tiled version fits comfortably within Kaggle's disk quota, enabling seamless dataset mounting and training.
- **OBB annotations:** DOTA provides oriented bounding box annotations with 4 corner points — perfectly matched to YOLOv8-OBB's input format.
- **Benchmark status:** DOTA v1.0 is the standard benchmark for aerial OBB detection, allowing our results to be contextualised against the published literature.
- **Bridge class included:** Bridge is one of DOTA's 15 classes, enabling direct bridge-specific AP evaluation.

3. DOTA v1.0 Dataset

DOTA (Dataset for Object Detection in Aerial Images) v1.0 is one of the largest and most widely used benchmarks for object detection in aerial imagery, published by researchers at Wuhan University (Xia et al., CVPR 2018). It has become the standard benchmark for oriented object detection in remote sensing.

Dataset source: https://www.kaggle.com/datasets/chandlertimm/dota-data

3.1 Dataset Overview

Property	Value
Total images	2,806 high-resolution aerial images (800 to 4,000+ pixels per side)
Annotations	188,282 instances across 15 object categories
Annotation format	Oriented Bounding Boxes (OBB) — 4 corner points (x1,y1,…,x4,y4)
Tile size used	Pre-tiled 1024×1024 px crops (Ultralytics DOTAv1 release)
Train / Val split	1,411 / 458 tiles used in this project

3.2 The 15 DOTA Classes

The dataset covers: plane, ship, storage-tank, baseball-diamond, tennis-court, basketball-court, ground-track-field, harbor, **bridge**, large-vehicle, small-vehicle, helicopter, roundabout, soccer-ball-field, swimming-pool.

3.3 Bridge Statistics in DOTA v1.0

Split	Total Images	Bridge Instances	Images w/ Bridge
Training Set	1,411	2,047	210 (14.9%)
Validation Set	458	464	75 (16.4%)
Total	1,869	2,511	285 tiles

Key challenge: only 14.9% of training tiles contain any bridge annotation. This severe class imbalance — vehicles and ships appear in nearly every tile while bridges appear in fewer than 1 in 7 — directly contributes to lower bridge-specific AP.

4. Methodology

4.1 Model Architecture — YOLOv8-OBB

YOLO (You Only Look Once) is a family of single-stage object detectors that predict bounding boxes and class probabilities directly from full images in a single forward pass. YOLOv8, released by Ultralytics in 2023, introduces an anchor-free detection head, a new C2f backbone module, and native support for Oriented Bounding Box (OBB) detection.

The OBB variant outputs rotated bounding boxes in the form **(x_center, y_center, width, height, angle)**, where angle is the rotation of the box. This is critical for aerial objects such as bridges that have no canonical orientation. The model uses a CSPDarknet-inspired backbone, a PANet feature pyramid neck, and a decoupled classification / regression detection head with an additional angle prediction branch.

Model Variants

Variant	Parameters	GFLOPs	Role
YOLOv8n-OBB (Nano)	3.1 M	8.4	Baseline — fast, small memory footprint
YOLOv8s-OBB (Small)	11.4 M	29.4	Improved — 3.7× capacity, better thin-object detection

4.2 Experiment Design

Two sequential experiments were conducted. The baseline established a performance floor using the smallest available OBB model. The improved experiment addressed identified weaknesses through a more capable model, extended training, more aggressive augmentation, and longer warmup.

- **Experiment 1 (Baseline):** YOLOv8n-OBB trained for 50 epochs with patience=10 and standard augmentation.
- **Experiment 2 (Improved):** YOLOv8s-OBB trained for 100 epochs with patience=20, copy-paste augmentation (p=0.3), extended warmup (5 epochs), and scale=0.6.

4.3 Hyperparameter Configuration

Parameter	Baseline (n)	Improved (s)
Model	YOLOv8n-OBB	YOLOv8s-OBB
Epochs	50	100
Image Size	1024×1024	1024×1024
Batch Size	8	8
Patience (Early Stop)	10	20
Warmup Epochs	3 (default)	5
Rotation (degrees)	180°	180°
Flip Up/Down	0.5	0.5
Flip Left/Right	0.5	0.5
Mosaic	1.0	1.0
Scale	0.5	0.6

Parameter	Baseline (n)	Improved (s)
Copy-Paste	0.0	0.3
Translate	0.1	0.1

4.4 Augmentation Strategy

Augmentation is critical for bridge detection because the training set has limited bridge examples and bridges can appear at any orientation. The following strategy was applied:

- **Full 360° rotation (degrees=180.0):** Bridges appear at arbitrary angles in aerial imagery. Full rotation ensures the model learns rotation-invariant features.
- **Mosaic augmentation (mosaic=1.0):** Combines 4 random tiles into a single training sample, increasing scene diversity and forcing the model to detect bridges among varied clutter.
- **Copy-Paste (copy_paste=0.3, Improved only):** Randomly copies bridge instances from other tiles and pastes them into the current tile. Directly addresses bridge class imbalance by artificially increasing bridge instance frequency.
- **Vertical / horizontal flip (flipud=0.5, fliplr=0.5):** Doubles effective dataset size for symmetric augmentation.
- **Scale jitter (scale=0.5–0.6):** Simulates bridges at different altitudes and zoom levels.
- **Translation (translate=0.1):** Reduces positional overfitting.

5. Results & Evaluation

5.1 Quantitative Results — Model Comparison

Metric	Baseline (YOLOv8n)	Improved (YOLOv8s)	Change	Significance
Overall mAP@50	0.516	0.549	+3.3%	Moderate
Overall mAP@50-95	0.340	0.371	+3.1%	Moderate
Bridge AP@50	0.163	0.192	+2.9%	Low — needs work
Precision	0.732	0.783	+5.1%	Good
Recall	0.503	0.528	+2.5%	Moderate
Mean Bridge Conf	0.231	0.293	+0.062	Meaningful
Bridge Dets >0.5	18	33	+83%	Significant

5.2 Key Performance Metrics (Best Model — YOLOv8s-OBB)

54.9%	37.1%	19.2%	78.3%	52.8%
mAP@50 (All Classes)	mAP@50-95 (All Classes)	Bridge AP@50	Precision	Recall

Metrics above are for the best-performing model (YOLOv8s-OBB) evaluated on the DOTA v1.0 validation set of 458 tiles (75 of which contain bridge annotations). Evaluation uses the standard COCO-style mAP protocol.

5.3 Bridge-Specific Confidence Analysis

A detailed confidence distribution analysis was carried out on 100 validation tiles using a deliberately low confidence threshold of 0.10, to capture even weak bridge detections and assess the models' detection tendencies comprehensively.

Metric	YOLOv8n (Baseline)	YOLOv8s (Improved)	Delta
Total bridge detections (conf > 0.1)	224	190	–34 (fewer FP)
Mean confidence	0.231	0.293	+0.062
High-confidence detections (> 0.5)	18	33	+83% ↑
Detection density per tile	2.24	1.90	More selective

The improved model produces fewer total detections but with higher mean confidence — indicating that the larger backbone is more selective. The 83% increase in high-confidence bridge detections (18 → 33) is the most meaningful indicator of practical improvement.

6. Discussion

6.1 Why Is Bridge AP@50 Low? (Root Cause Analysis)

The bridge AP@50 of 19.2% is the most critically discussed result. Bridge is consistently one of the lowest-performing classes even in published DOTA research, and the reasons are well-understood:

- **Severe class imbalance:** Bridges account for approximately 2,047 of ~100,000+ total DOTA instances (<2%). The model is overwhelmed by vehicles and ships which dominate gradient updates during training.
- **Extreme aspect ratios:** Bridges are long and thin. IoU-based OBB loss penalises imprecise aspect ratio predictions heavily, making it harder to accumulate high-IoU matches.
- **Visual similarity to roads:** At tile resolution, bridges and roads are structurally identical — both are elongated, straight, grey/black. Without water context, discrimination is very difficult.
- **Low-resolution features:** At 1024 px input with the small model's stride-32 feature map, a narrow bridge may only span 2–4 feature map cells in width, leaving very few pixels for discriminative feature extraction.

6.2 Improvements Achieved Over Baseline

Despite the challenges above, the improved YOLOv8s-OBB model demonstrates clear progress over the baseline across every measured metric:

- Overall mAP@50 improved from 51.6% to 54.9% — a meaningful 3.3 pp gain attributable to the larger backbone's feature extraction capacity.
- Bridge AP@50 improved from 16.3% to 19.2%, representing an **18% relative improvement**. Given the difficulty of the class, this is a substantive gain.
- Precision improved from 73.2% to 78.3%, meaning fewer false positive detections across all classes.
- High-confidence bridge detections (conf > 0.5) increased from 18 to 33 — an **83% increase**, indicating the model is substantially more certain when it does detect a bridge.
- Extended training (100 vs 50 epochs) and higher patience (20 vs 10) ensured the model was not stopped prematurely and had time to converge on rare class features.
- Copy-paste augmentation (p=0.3) directly addresses bridge rarity by synthesising additional bridge examples into non-bridge tiles.

7. Limitations & Future Work

7.1 Current Limitations

- **Limited bridge diversity:** DOTA is geographically biased toward Chinese urban areas. Bridge types, materials, sizes, and water body characteristics in other geographic regions are underrepresented.
- **No temporal analysis:** The system is a single-frame detector. Change detection (bridge construction, damage, collapse) would require multi-temporal imagery.
- **Single-resolution inference:** All inference is performed at a fixed 1024×1024 tile size. Multi-scale test-time augmentation (TTA) could improve recall for small bridges.
- **Small model capacity:** YOLOv8s is still a relatively compact model. Larger backbones (YOLOv8m, YOLOv8l, or RT-DETR) are expected to provide significant AP gains.
- **GPU/disk constraints:** As encountered with the GLH-Bridge dataset, GPU memory and cloud disk quotas remain a practical bottleneck for scaling to higher-resolution datasets.

7.2 Proposed Future Improvements

Priority	Improvement	Expected Gain
High	Upgrade to YOLOv8m-OBB or YOLOv8l-OBB	+5–8% absolute mAP
High	Two-stage water-body conditioned detection (SegFormer-B0 mask)	Fewer FP, focused compute
Medium	Bridge-class weighted loss (4× bridge vs. vehicle weight)	Higher bridge AP
Medium	Supplementary dataset integration (FAIR1M, xView/xBD)	Better generalisation
Medium	Multi-scale inference with tile overlap (20%) + NMS	Better recall at boundaries
Low	Revisit GLH-Bridge with proper HPC infrastructure (A100 GPU + large disk)	Higher-res bridge features

8. Conclusion

This project successfully developed and evaluated an end-to-end deep learning pipeline for detecting bridges over water bodies in aerial imagery using the YOLOv8-OBB framework on the DOTA v1.0 dataset. The work demonstrates the complete ML workflow from dataset acquisition and verification through model training, evaluation, and deployable inference.

An important practical contribution of this project is the documented dataset selection process. The initial attempt to use the GLH-Bridge dataset — a purpose-built, high-resolution bridge dataset — was abandoned due to GPU memory exhaustion, CUDA runtime errors, and a ~50 GB dataset size that exceeded Kaggle and Colab disk quotas. This experience motivated a principled pivot to DOTA v1.0, whose pre-tiled 1024×1024 format is GPU-ready and represents the standard benchmark for aerial OBB detection.

The improved YOLOv8s-OBB model achieves an overall mAP@50 of 54.9% on the DOTA v1.0 validation set, with a Bridge-specific AP@50 of 19.2%. While the bridge AP is modest in absolute terms, the analysis clearly demonstrates that this is driven by fundamental dataset characteristics (class rarity, visual ambiguity, thin structure) rather than methodological failure. The systematic comparison between baseline and improved models shows consistent improvement across every metric, validating the design choices in the second experiment.

Key contributions of this project: (1) demonstrating the superiority of OBB detection over standard axis-aligned boxes for elongated aerial objects; (2) quantifying bridge class difficulty through confidence distribution analysis; (3) identifying copy-paste augmentation as the most effective technique for rare-class improvement; (4) delivering a reusable, well-documented inference module for real-world deployment; and (5) providing a clear, reproducible account of the dataset engineering challenges encountered.

Future work should prioritise a larger model backbone, water-body semantic conditioning, and class-weighted loss to push bridge AP into a more operationally useful range (>50%). The foundation built in this project provides a solid basis for these extensions.

9. References & Tools

Dataset

Xia, G.S., et al. (2018). DOTA: A Large-Scale Dataset for Object Detection in Aerial Images. *CVPR 2018*. Available at: captain-whu.github.io/DOTA

Ultralytics DOTAv1 pre-tiled release: github.com/ultralytics/assets — 1024×1024 pre-tiled version used in this project.

GLH-Bridge Dataset (initial attempt): 6,000 high-resolution bridge images (2,048×2,048 to 16,384×16,384 px). Available at: pan.baidu.com (GLH-Bridge). Not used in final experiments due to GPU/disk constraints.

Model & Framework

Jocher, G., et al. (2023). Ultralytics YOLOv8. Ultralytics. docs.ultralytics.com

Redmon, J., & Farhadi, A. (2018). YOLOv3: An Incremental Improvement. arXiv:1804.02767.

Bochkovskiy, A., Wang, C.Y., & Liao, H.Y.M. (2020). YOLOv4: Optimal Speed and Accuracy. arXiv:2004.10934.

Tools & Infrastructure

Tool / Library	Version	Purpose
PyTorch	2.1.0 + CUDA 13.0	Deep learning backend
Ultralytics	8.4.37	YOLOv8 training and inference library
OpenCV	4.x	Image loading and visualisation
Kaggle Notebooks	Tesla T4 GPU, 16 GB VRAM	Training platform
Matplotlib	—	Training curves, detection visualisation
NumPy / Pandas	—	Numerical analysis and results CSV parsing

End of Report